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**A Hybrid Model for Multiscale Laser
Plasma Simulations with Detailed
Collisional Physics**



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1.0 SUMMARY

The focus of this effort was to improve the knowledge gained through spectroscopy and improve the efficiency of the algorithms used to compute the spectra from plasma. To this end, an Air Force Office of Scientific (AFOSR) project was submitted and awarded to investigate this. Two primary tasks were performed under this effort, one was to improve the level grouping technique for collisional radiative (CR) simulations and the second was the development of an algorithm to determine the electron energy distribution function (eEDF) from optical emission spectroscopy.

Level grouping is a complexity reduction technique applied to detailed collisional radiative (CR) calculations wherein multiple excited states populate into a single group and a relationship is prescribed that defines the density relationship between the levels. In prior work, the electronic states grouping was done manually and required an iterative approach to find an optimum grouping. This work uses spectral clustering to automate and optimize the groups. This approach was successful, providing similar steady state results as the detailed model. Future work on this model was planned but not executed due to lack of personnel.

The eEDF inference problem was quite difficult and required solving an “inverse problem” wherein the final solution is known and the initial conditions that result in measured spectra must be determined. A Bayesian Markov Chain Monte Carlo (MCMC) approach was used to vary the eEDF and the resulting spectra was calculated. A sample problem was created, wherein an eEDF was prescribed and the spectra was calculated. Then several candidate eEDFs were selected through the MCMC method and compared to the reference spectra. The resulting eEDFs were clustered around the prescribed eEDF and further work was planned but not executed due to lack of personnel.

2.0 INTRODUCTION

This work unit (WU) effort contained the work completed under an AFOSR Laboratory Research Initiation Request (LRIR), “Accelerated Models for Transient Plasmas” funded by the Plasma and Electroenergetic Physics portfolio. The overall goal of the AFOSR LRIR was to accelerate the algorithms used for CR simulations and apply them in areas to aid in the development and understanding of high-power electric propulsion (EP) devices. To this end, two key areas of research were performed under this WU: the automation of the grouping algorithm for the Boltzmann grouping assumption for CR calculations and the development of a method to determine the electron EDF from spectra.

The automatic grouping algorithm was an extension of the work done on a prior LRIR where the creation of an optimal group was identified as an area of future work. When performing CR calculations, one of the assumptions that must be made is how to represent the distribution of electrons in a molecule or atom, i.e., the electron excited state population, this is referred to as the atomic state distribution function (ASDF). In a detailed or full model every excited state, up to a particular energy, is solved for as an independent variable. On the other extreme, is the Quasi Steady State (QSS) approach that only solves for the ground state density and uses a series of steady state assumptions to determine the population of the excited states. A third approach is to group excited states and assume that the electrons in the excited states move in a like manner. The approach developed under a previous WU, used the grouping technique and assumed that the population in a group was defined by a Boltzmann distribution at a set temperature. The focus of the work in this effort was to develop a method to automate and optimize the groups. This report will summarize the findings for this effort, for a full account please see the journal article by Abrantes et al. [1].

The second project seeks to infer the eEDF from the emission or absorption spectra. A standard assumption when analyzing the spectra from a plasma is that the electrons follow a Maxwellian distribution function. In general, this is an accurate assumption to make due to the high frequency of electron-electron collisions driving the distribution function to a Maxwellian. But the presence of magnetic fields or strong impact transitions can drive the steady state distribution away from Maxwellian. This work developed new methods to represent the eEDF, as well as a technique to solve the inverse problem. This report will summarize the findings for this effort, for a full account please see the Technical Report by Cambier et al. [2]¹.

¹ Report is distro A, and can be found on DTIC: <https://apps.dtic.mil/sti/citations/AD1173560>

3.0 METHODS, ASSUMPTIONS, AND PROCEDURES

Automatic Grouping Algorithm

The atomic grouping algorithm uses spectral clustering to automate and optimize the groups, an approach developed by Sahai et al. [3]. The first step in this approach was to define a connection matrix that represented the strength of the connectedness between atomic levels. The next step was to feed the connection matrix into a Gaussian similarity function producing a graph adjacency matrix. The third step calculated the asymmetric normalized graph Laplacian then extract the eigenvectors that correspond to the smallest eigenvalues of the graph Laplacian. The eigenvectors were inserted into a new matrix as columns and a conventional clustering algorithm is applied over its rows. An example of the grouping is shown in Figure 1. In this case, the first five levels for each atom composed the first group, after that, the number of levels in each group contained a few dozen to over a hundred levels.

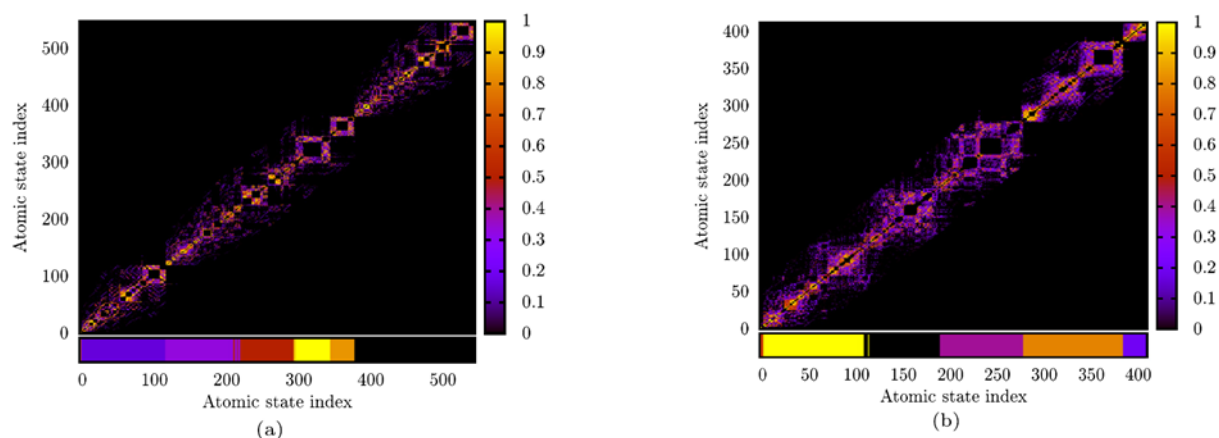


Figure 1. Gaussian Similarity Matrix and Clustering for Atomic Oxygen (a), and Atomic Oxygen Ion (b). Vertical Color Bar is the Connectedness, and the Horizontal Color Bar are the Groups

Plots a & b from Figure 1 of Abrantes et al. [1]

eEDF Inference

There are two methods to the eEDF inference project, the first is the representation of the eEDF, and the second is a method used to explore the different eEDF configurations. The goal in representing the eEDF was to reduce its dimensionality while maintaining certain properties, such as positivity and zero (or near zero) values at the boundaries. A robust approach was to represent the distribution function as a cumulative distribution function (CDF), instead of a probability distribution function. The first approach used linear splines, which guaranteed positivity, monotonicity, and had a single degree of freedom per control point but required many control points to accurately represent a PDF and did not provide zero weight at the boundaries. The second approach used was to represent the CDF as an exponential spline that had diminishing derivatives at the boundaries. For added accuracy points were selected in log space, increasing the number of points in the high energy tail.

Two approaches were used to explore the trade space, the Transitional Markov chain Monte Carlo (TMCMC) technique and kernel-embedding (KE) conditional distributions. Monte

Carlo methods work by randomly sampling the trade space and in cases of high dimensionality and non-linearity it may be the only method that is computationally feasible. The downside of MC methods is that the randomness of the algorithm leads to slow convergence. The TMCMC method adapts the distribution for each degree of freedom, allowing it to better capture multimodal distributions [4]. Kernel embedding conditional distributions is a process wherein the kernel embedding is used to simplify the relationships between variables, typically requiring an elevation of the dimensionality, and conditional distributions. The KE work was largely based on the work of Song et al. [5] and Fukumizu et al. [6].

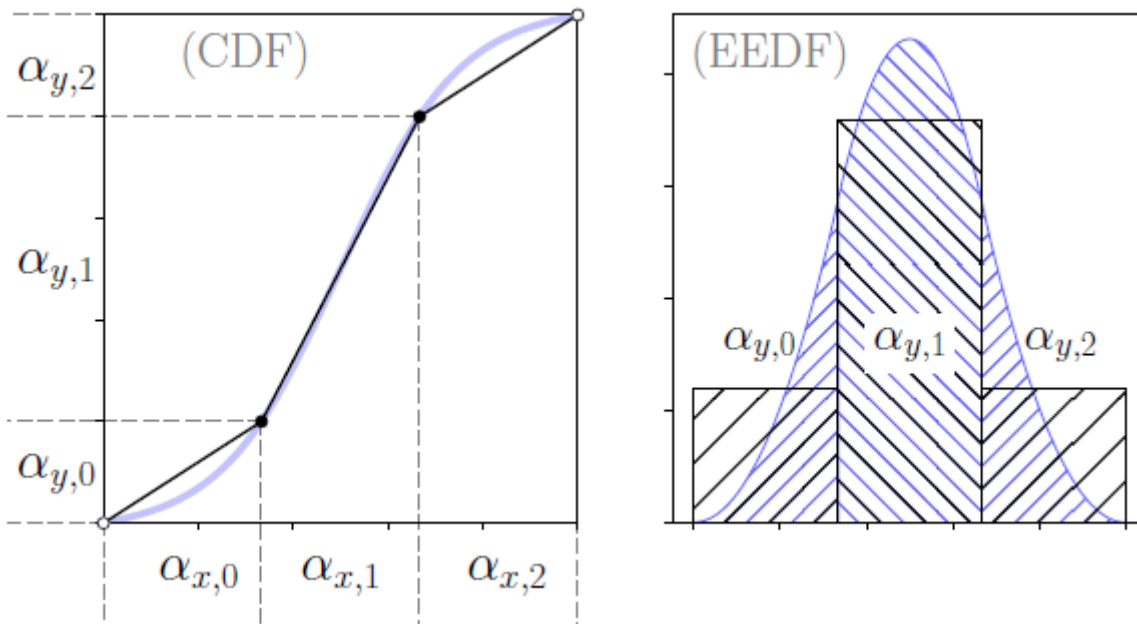


Figure 2. Left Plot Shows Piecewise-Linear (Thin, Black) and Exponential-Tension Spline (Thick Blue) Interpolation of a Maxwellian CDF, Right Plot is the Corresponding PDF

This is Figure 1 from [2]

4.0 RESULTS AND DISCUSSION

Automatic Grouping Algorithm

To test the automatic clustering for the complexity reduction collisional radiative model, atomic oxygen was reacted with an electron heat bath at 10 eV. Two different cases were run, a collisional (10^{24} #/m³) and coronal (10^{20} #/m³) case. For comparison, the complexity reduction technique was compared to a detailed model (i.e., every excited state is explicitly modeled). The results, shown in Figure 3, show that the complexity reduction with automatic grouping compares favorably with the full model. For the coronal case, on the right side of Figure 3, the CR method compares well with low energies and for neutral atomic oxygen but has difficulties capturing the higher energy shells for singly and double charged atomic oxygen. The largest impact that the different ASDFs had is that the spectra will not be the same and certain lines cannot be used to validate the model. Another potential impact is to the total emitted radiation. A comparison between the full and reduced CR model is shown in Figure 4, in this figure there is little difference in the steady state radiation rate between the two models. Although there are differences in the unsteady regions, the radiation emitted during this time is miniscule, so the impact, from a design perspective, is minimal. The coronal and collisional cases resulted in an approximate 7x speed up while reducing the number of variables from 1,143 to 36, or about a 32x decrease.

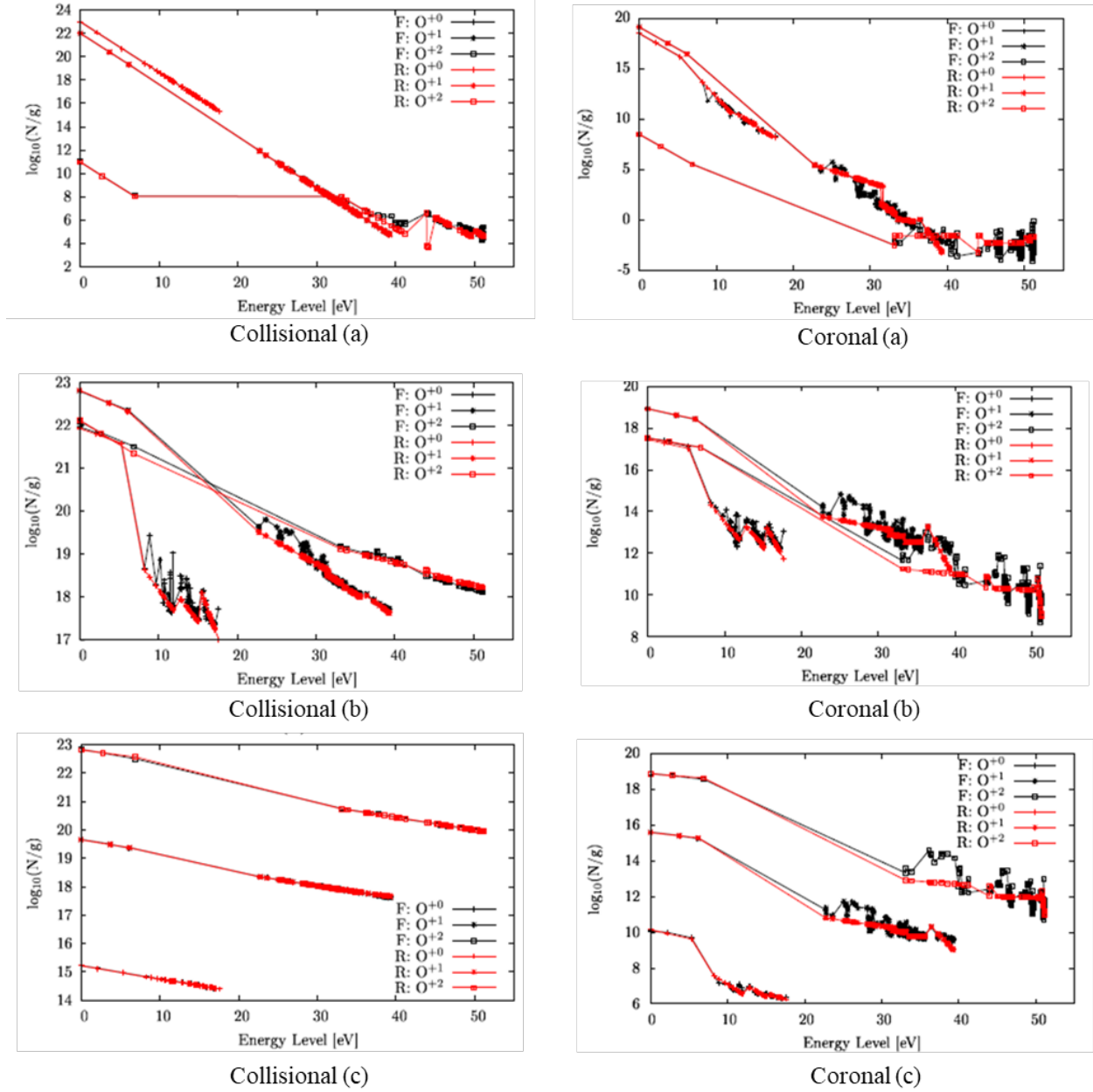


Figure 3. Plots on Left Hand Side are Related to the Collisional Case at Times 0.0, 1.82×10^{-10} , and 1.5×10^{-9} (Steady State) for A, B, and C Respectively. Plots on Right Hand Side are From the Coronal Cases at Times 0.0, 1.08×10^{-6} , and 1.00×10^{-4} (Steady State) Seconds for A, B, C Respectively

Figure Adapted from Figures 2 and 6 from Abrantes et al [1]

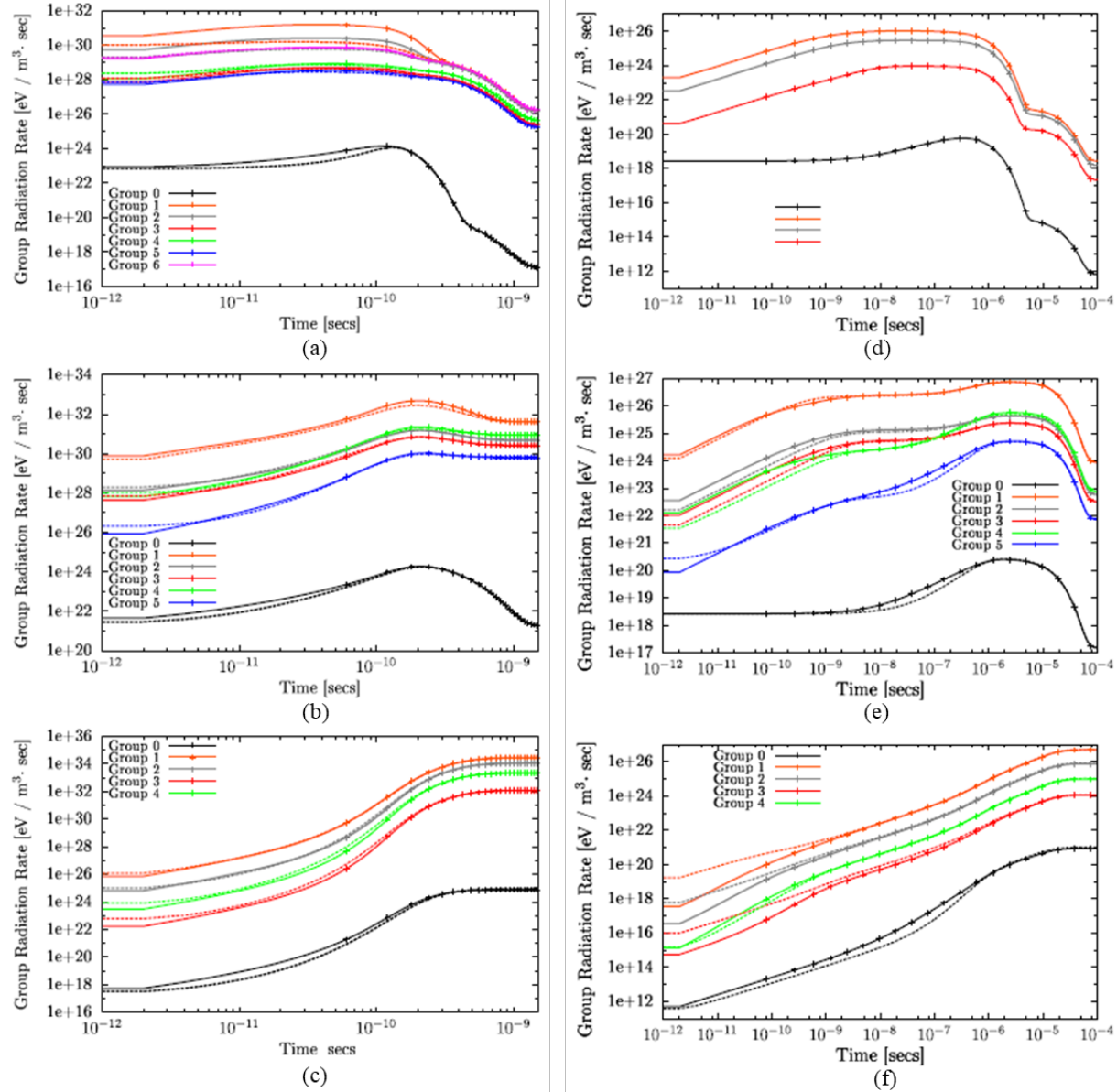


Figure 4. Comparison of Radiation Losses for Bound-Bound Transitions for Each Group for O, O⁺, And O⁺⁺, Top, Middle, and Bottom Respectively. Plots on Left are for Collisional Case and Right are for Coronal. Full Model Shown as Solid and Reduced are Dashed

From Figures 4 and 7 from Abrantes et al. [1]

eEDF Inference

To test the eEDF inference algorithm, an eEDF was assumed and a corresponding observable was generated. In this study, the emitted spectra were not used as an observable, instead the population at seven different levels were used. Matching the spectra was too difficult for the initial investigation and as spectra cannot be matched without first matching the populations, making this observable relevant to the end goal. Example outputs are shown in

Figure 5 and Figure 6, for the TMCMC and KE methods respectively. The exponential spline representation of the TMCMC method, (seen in the left plot of Figure 5), used four bins and as expected with exponential fits, did a good job representing EDFs that were Maxwellian like. Increasing the number of bins causes oscillations. The I-splines representation of the TMCMC method, (seen in the right plot of Figure 5) provided more varied results and did not converge as well as the exponential spline representation.

The Kernel-embedded method results are shown in Figure 6. In this plot, the axis is zoomed into the low energy region and shows the method's attempt to capture non-Maxwell Boltzmann distribution features. The training data used for this method splines to match a Maxwell Boltzmann (MB) distribution and then applied perturbations to these values providing non-Maxwellian eEDFs. Figure 7 shows how well the method was able to match the target observable: circles are the true values, and the lines represent different possible solutions. One of the usable defined parameters is the weighting between the MB curve and the perturbations. Figure 8 shows an example where the MB had a weight of 90%, this approach leads to more clustering around the target solution and provides an overall better fit than the initial KE approach shown in Figure 6.

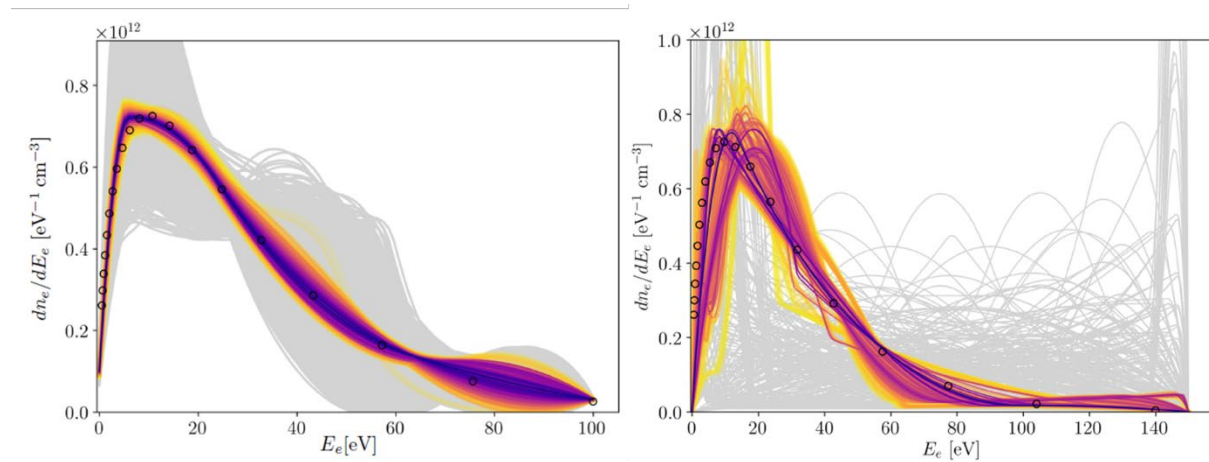


Figure 5. Example Using TMCMC for Exponential Tension-Splines (left) and I-Splines CDF Basis Functions (right). The Background Gray Lines Represent Initial Grids, the Circles are the Target eEDF, and the Colored Lines are the Possible eEDFs Where the Darker Lines Represent More Probable eEDFs.

Figures 3 & 4 from Cambier et al. [2]

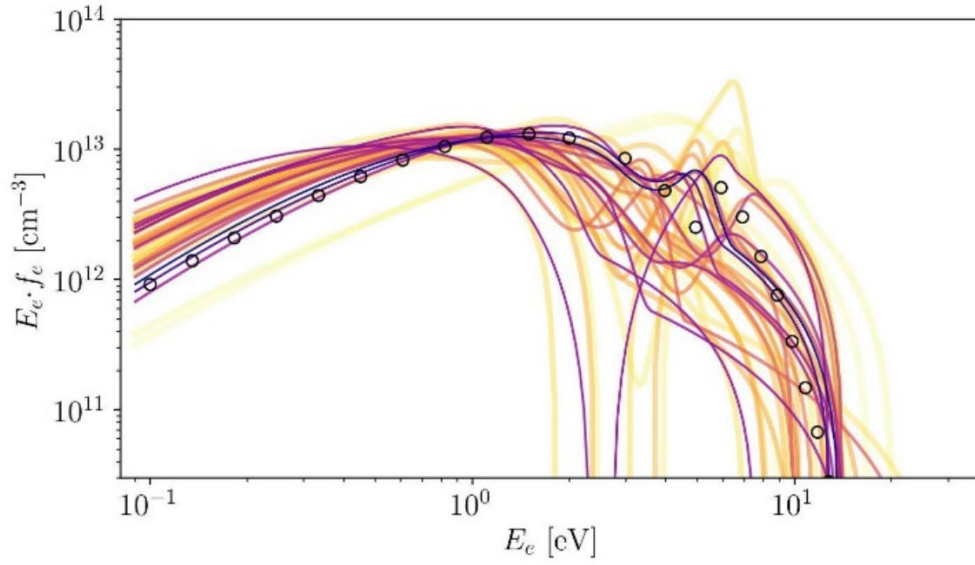


Figure 6. Example Using the Kernel-Embedded Method, Color Scheme is the Same as Used in Figure 5

This is Figure 6 from [2]

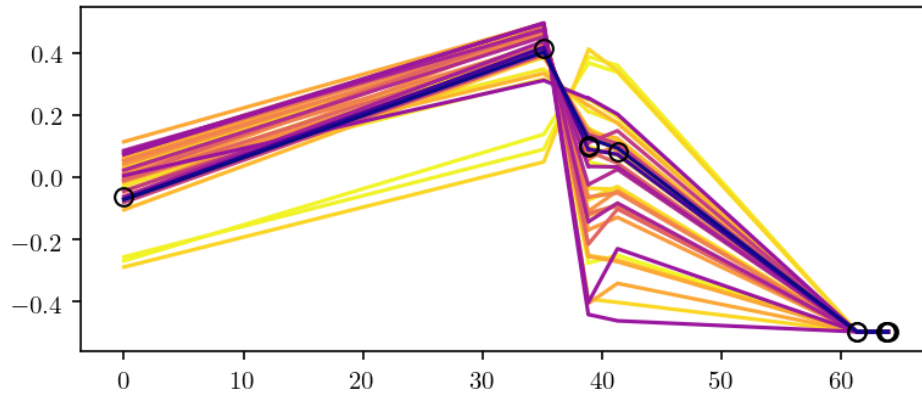


Figure 7. Transformed Electron Level Population, Serves as the Observable to the eEDFs Shown in Figure 6

Figure 7 in Cambier et al. [2]

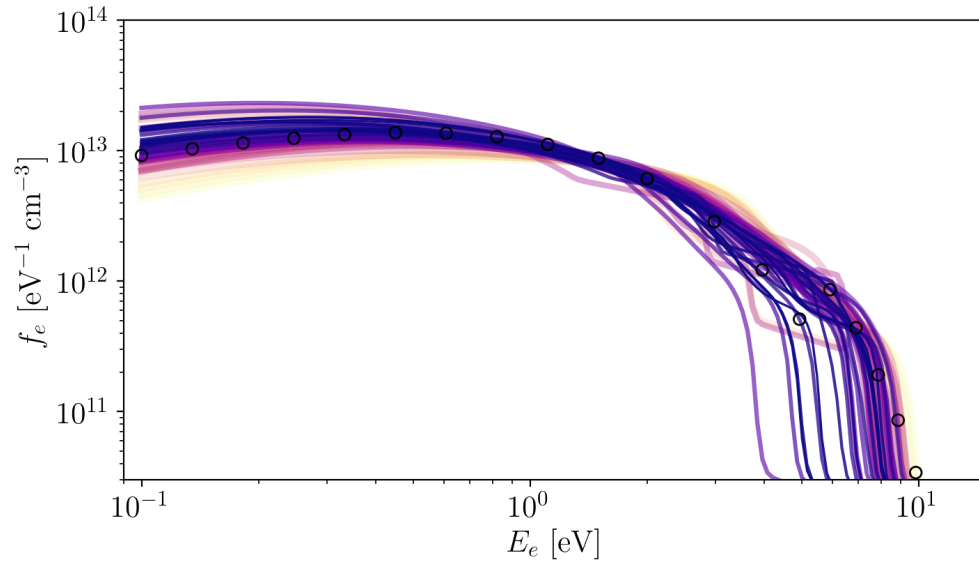


Figure 8. Example Using Kernel-Embedding With A Maxwell Boltzmann Prior

Figure 12 in Cambier et al. [2]

5.0 CONCLUSIONS

In conclusion, this WU oversaw the development of two different models. The first was the automatic group formation in a reduced CR model and the second was the eEDF inference model. The complexity reduction technique addressed an issue with the initial method that manually adjusted the groups to provide both a more effective solution that balanced accuracy and computational cost. The reduced model compared favorably with the full model for the collisional case and matched the low energy populations and the steady state radiation rate for the coronal case but did not properly capture the high energy population levels and some transient properties.

In the eEDF inference project, two different methods were used to infer the eEDF required to produce the target electron level populations. The TMCMC method is a more traditional choice where the eEDF was parameterized and represented by different types of splines, whose properties were randomized through a MCMC process. This approach led to a high degree of clustering around the desired profile, but the most probable curve couldn't fit all the points. The second approach used a kernel-embedding method that was trained on several reference eEDFs that were allowed to deviate from a MB distribution. The KE approach did not converge as well as the TMCMC approach but was still able to follow the general trend of the target eEDF.

6.0 RECOMMENDATIONS

Both automatic grouping and the eEDF inference projects showed great promise in the goal of better diagnosing and understanding of electric propulsion devices. The automatic grouping can be improved by extending the validation to more realistic conditions, different clustering criteria, and an adaptive clustering that varies in time. The eEDF inference project was a difficult task and could benefit from including anisotropic conditions, improvements to the algorithm, and extension to more realistic conditions.

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LIST OF SYMBOLS, ABBREVIATIONS AND ACRONYMS

AFOSR	Air Force Office of Scientific Research
ASDF	atomic state distribution function
CDF	cumulative distribution function
CR	collisional radiative
eEDF	electron energy distribution function
EDF	energy distribution function
EP	Electric Propulsion
KE	kernel-embedding
LRIR	laboratory research initiation request
MB	Maxwell Boltzmann
PDF	Portable Document Format
QSS	quasi steady state
TMCMC	transitional Markov chain Monte Carlo
WU	Work unit